

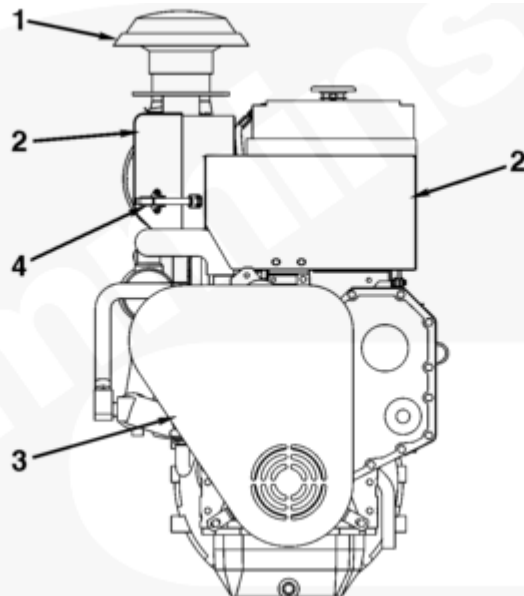
Trainings ▾

Engine Views

The following illustrations show the locations of the major external engine components, filters, and other service and maintenance points. Some external components will be at different locations for different engine models.

Note : The illustrations are **only** a reference to show a typical engine.

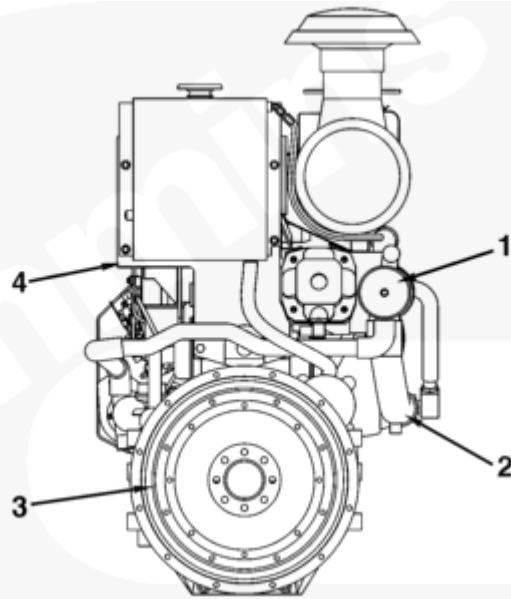
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00900448

6BTA5.9-DM Front View

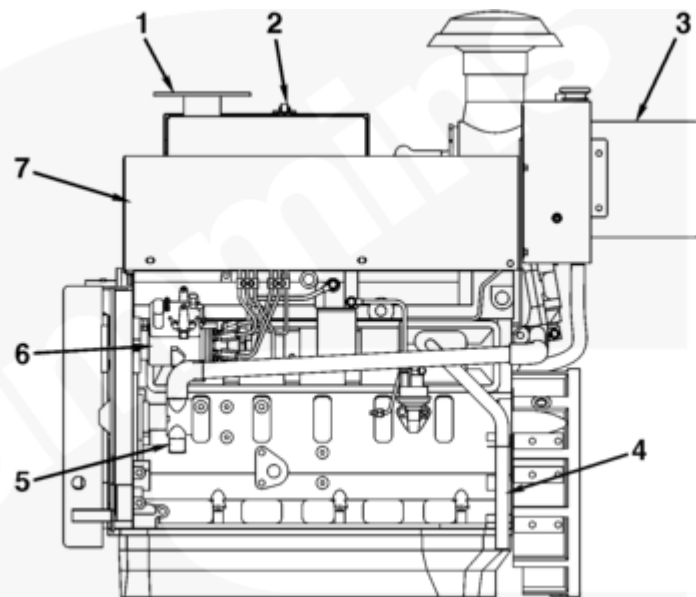
- 1. Air cleaner inlet
- 2. Heat shield
- 3. Belt and damper guard
- 4. Heat shield connector clamp



00900449

6BTA5.9-DM Rear View

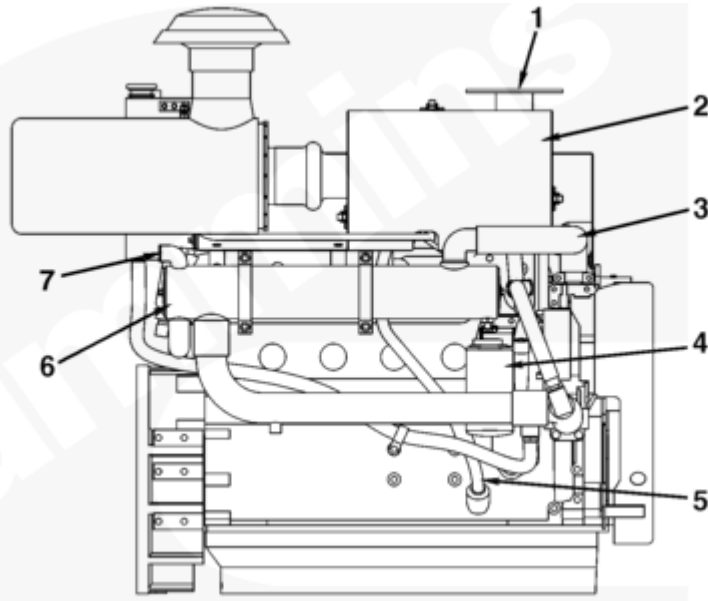
1. Heat exchanger
2. Engine coolant out of heat exchanger
3. Flywheel
4. Expansion tank mounting bracket



00900450

6BTA5.9-DM Fuel Pump Side

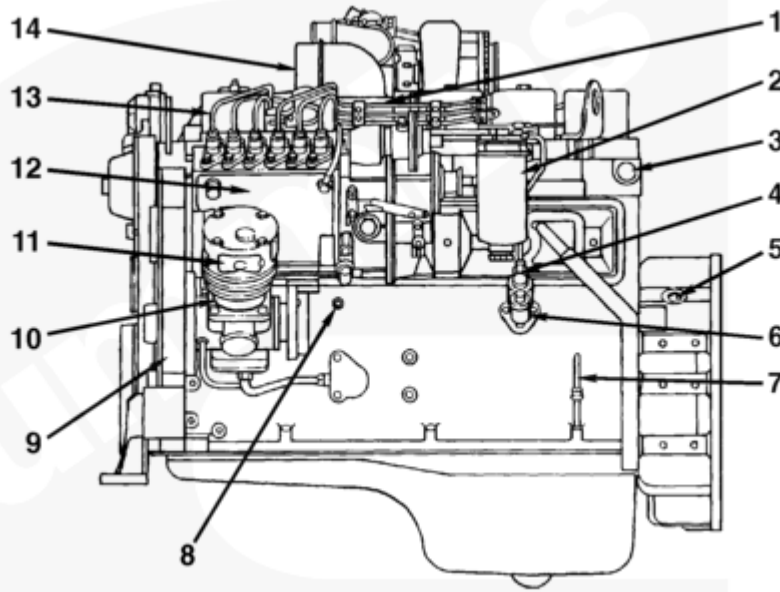
1. Exhaust outlet
2. Heat shield connector clamps
3. Air cleaner
4. Blowby outlet tube
5. Sea water inlet
6. Fuel pump
7. Heat shield



00900451

6BTA5.9-DM Exhaust Side

1. Exhaust outlet connection
2. Heat shield
3. Heat exchanger coolant inlet
4. Lubricating oil filter
5. Turbocharger oil drain connection
6. Heat exchanger
7. Heat exchanger sea water outlet



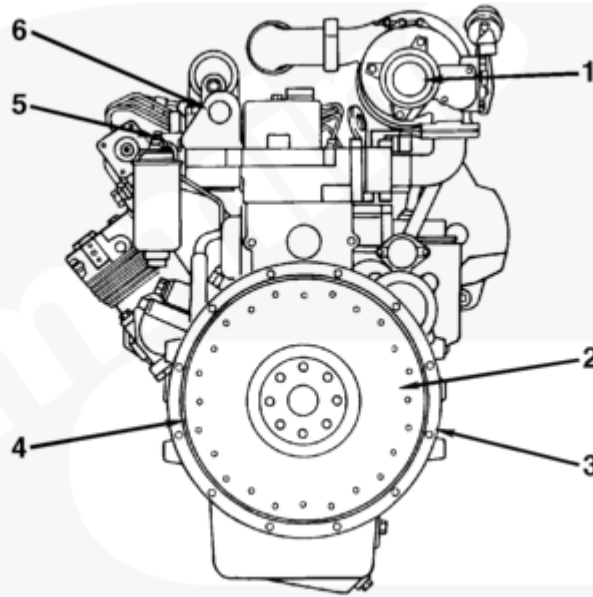
00900280

Automotive - Fuel Pump Side View

1. Intake air preheater (optional)
2. Fuel filter/water separator
3. Water heater supply (3/4-inch NPTF)
4. Fuel inlet connection (1/4-inch NPTF)

5. Magnetic pickup location (3/4-16 UNF)
6. Fuel lift pump
7. Dipstick
8. Oil pressure (1/8-inch NPTF)
9. Engine dataplate
10. Air compressor
11. Air compressor intake
12. In-line fuel injection pump
13. High pressure fuel lines
14. Engine air inlet.

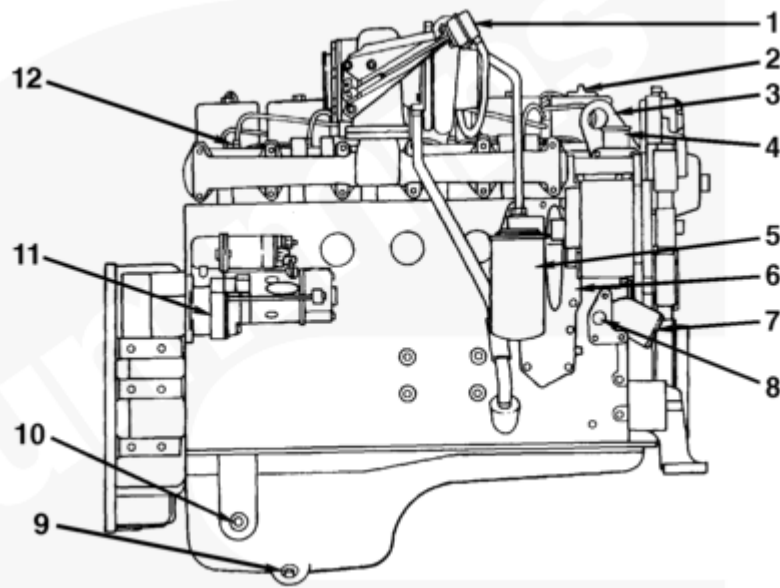
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00900279

Automotive - Rear View

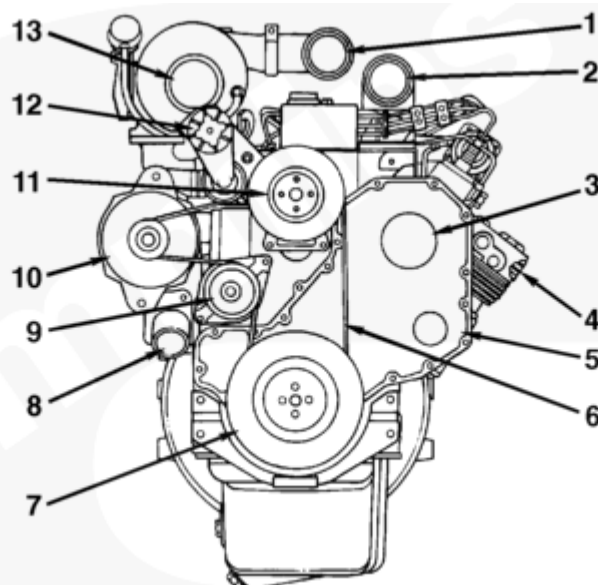
1. Turbocharger exhaust outlet
2. Clutch mounting plate
3. Flywheel housing
4. Flywheel flexplate
5. Fuel return connection
6. Rear engine lifting bracket.



00900278

Automotive - Turbocharger Side View

1. Turbocharger wastegate actuator
2. Oil fill
3. Front engine lifting bracket
4. Water outlet
5. Lubricating oil filter
6. Lubricating oil cooler
7. Water inlet
8. Water heater return (1/2-inch NPTF)
9. Oil drain
10. Provision for oil immersion heater
11. Starter motor solenoid
12. Fuel injection nozzles.

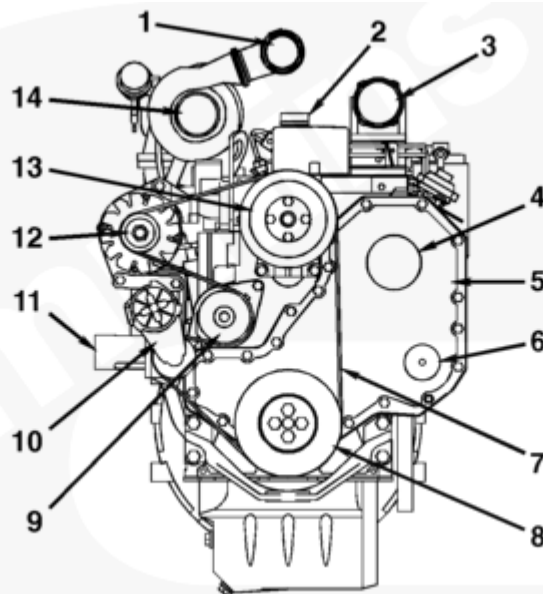


00900277

Automotive - Front View

1. Turbocharger air outlet
2. Engine air inlet
3. Fuel pump drive cover
4. Air compressor air outlet
5. Front gear cover
6. Fan drive belt
7. Vibration damper
8. Water inlet
9. Water pump
10. Alternator
11. Fan pulley
12. Automatic belt tensioner
13. Turbocharger air inlet.

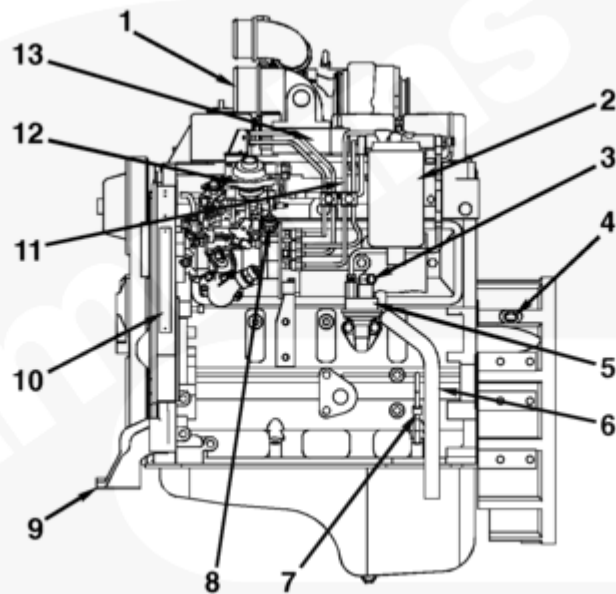
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00900346

Front View - B3.9 Industrial

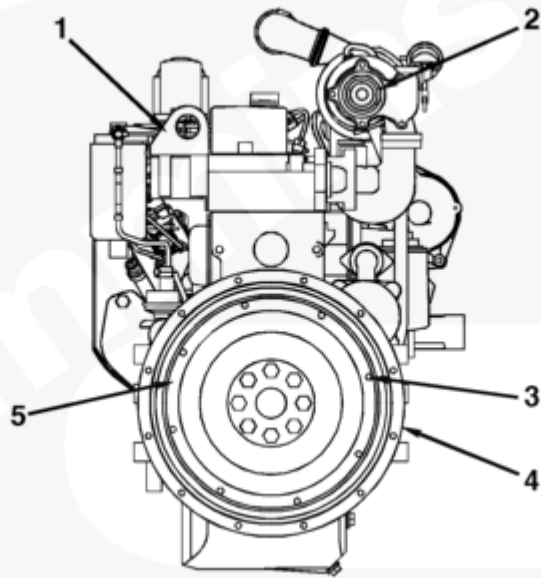
1. Turbocharger air outlet
2. Lubricating oil fill
3. Engine air inlet
4. Fuel pump drive cover
5. Front gear cover
6. Accessory drive cover (optional)
7. Fan drive belt
8. Front Pulley
9. Water pump
10. Automatic belt tensioner
11. Water inlet
12. Alternator
13. Fan pulley
14. Turbocharger air inlet.



00900343

Left Side View - B3.9 Industrial

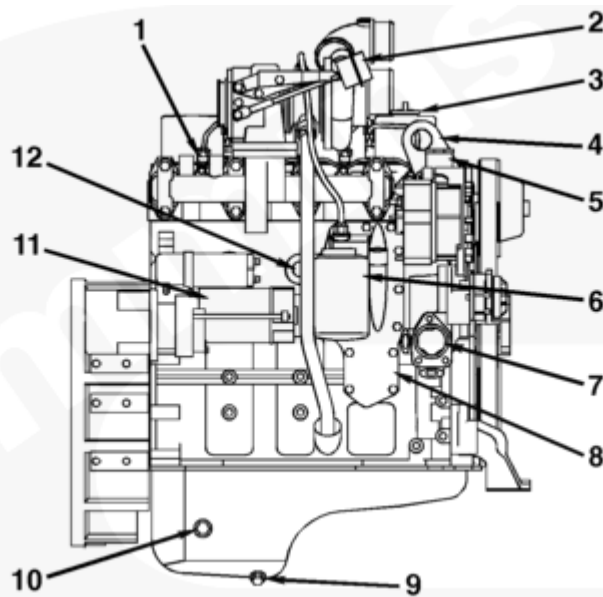
1. Engine air inlet
2. Fuel filter/water separator
3. Fuel inlet connection
4. Magnetic pickup location (3/4-16 UNF)
5. Fuel lift pump
6. Crankcase breather tube
7. Lubricating oil dipstick
8. Fuel drain connection
9. Front engine mounting bracket
10. Engine dataplate
11. High pressure fuel lines
12. Fuel injection pump
13. Intake air preheater (optional).



00900344

Rear View - B3.9 Industrial

1. Rear engine lifting bracket
2. Turbocharger exhaust outlet
3. Flexplate mounting holes
4. Flywheel housing
5. Flywheel/flexplate.



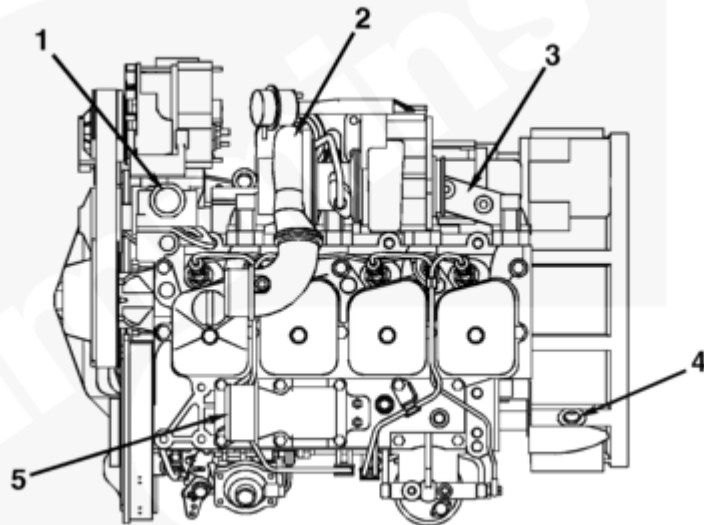
00900345

Right Side View - B3.9 Industrial

1. Fuel injection nozzles
2. Turbocharger wastegate actuator
3. Lubricating oil fill
4. Front engine lifting bracket
5. Water outlet
6. Lubricating oil filter

- 7. Water inlet
- 8. Lubricating oil cooler
- 9. Lubricating oil drain
- 10. Provision for lubricating oil immersion heater
- 11. Starter motor and solenoid
- 12. Provision for coolant heater.

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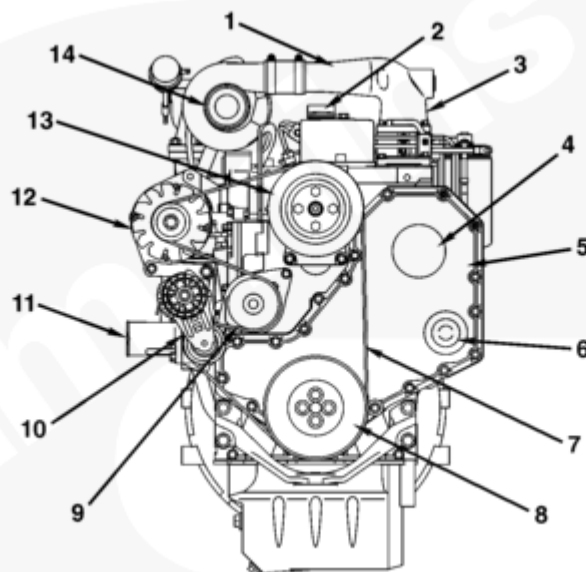


00900347

Top View - B3.9 Industrial

- 1. Water outlet
- 2. Turbocharger
- 3. Exhaust manifold
- 4. Magnetic pickup location (3/4-16 UNF)
- 5. Engine air inlet.

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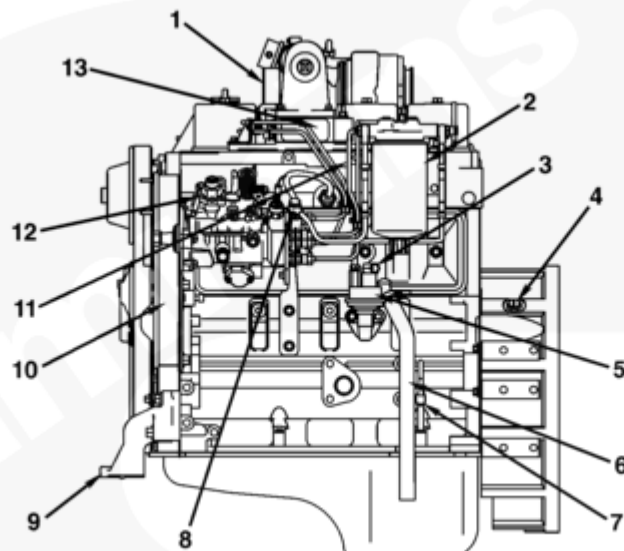


00900383

Front View - B4.5 Industrial

1. Turbocharger air outlet
2. Lubricating oil fill
3. Engine air inlet
4. Fuel pump drive cover
5. Front gear cover
6. Accessory drive cover (optional)
7. Fan drive belt
8. Front Pulley
9. Water pump
10. Automatic belt tensioner
11. Water inlet
12. Alternator
13. Fan pulley
14. Turbocharger air inlet.

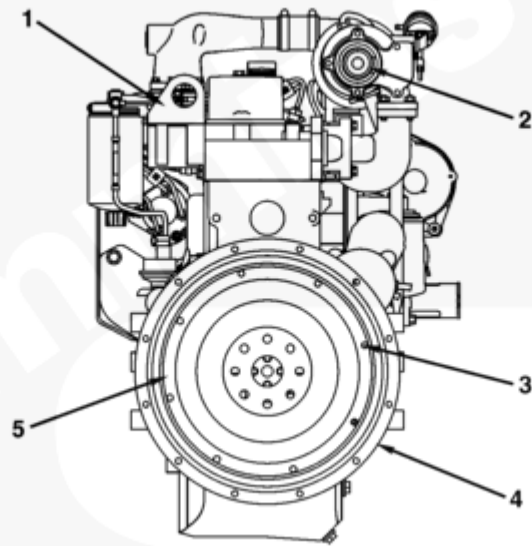
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Left Side View - B4.5 Industrial

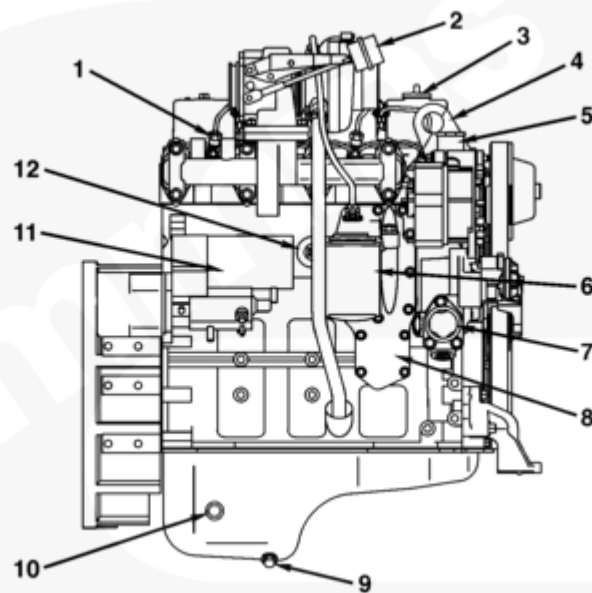
1. Engine air inlet
2. Fuel filter/water separator
3. Fuel inlet connection
4. Magnetic pickup location (3/4-16 UNF)
5. Fuel lift pump
6. Crankcase breather tube
7. Lubricating oil dipstick
8. Fuel drain connection
9. Front engine mounting bracket
10. Engine dataplate
11. High pressure fuel lines
12. Fuel injection pump
13. Intake air preheater (optional).



00900381

Rear View - B4.5 Industrial

1. Rear engine lifting bracket
2. Turbocharger exhaust outlet
3. Flexplate mounting holes
4. Flywheel housing
5. Flywheel/flexplate.



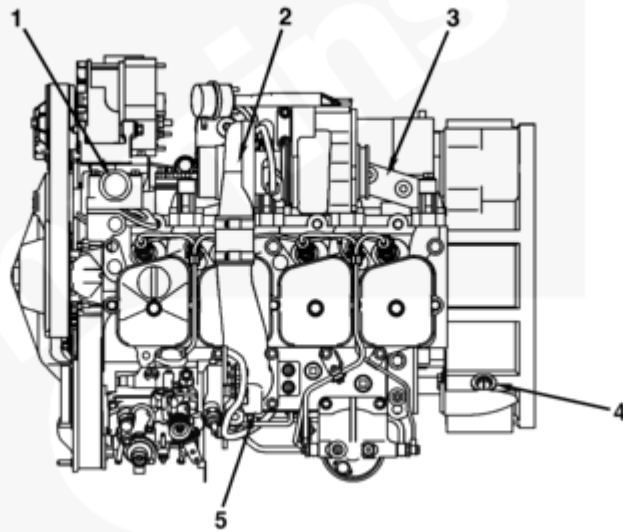
00900382

Right Side View - B4.5 Industrial

1. Fuel injection nozzles
2. Turbocharger wastegate actuator
3. Lubricating oil fill
4. Front engine lifting bracket
5. Water outlet
6. Lubricating oil filter

7. Water inlet
8. Lubricating oil cooler
9. Lubricating oil drain
10. Provision for lubricating oil immersion heater
11. Starter motor and solenoid
12. Provision for coolant heater.

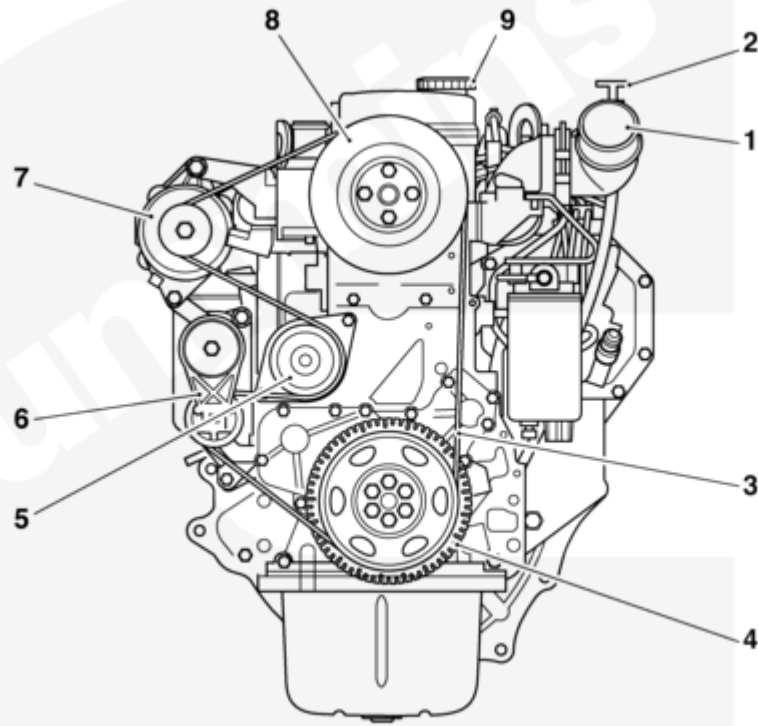
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00900384

Top View - B4.5 Industrial

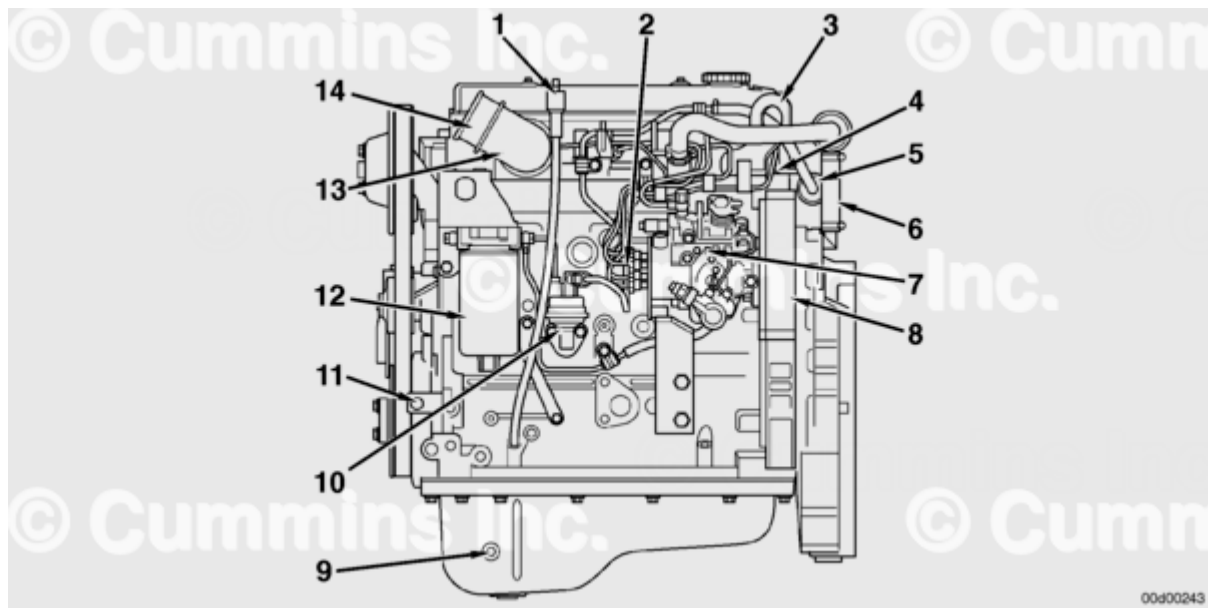
1. Water outlet
2. Turbocharger
3. Exhaust manifold
4. Magnetic pickup location (3/4-16 UNF)
5. Engine air inlet.



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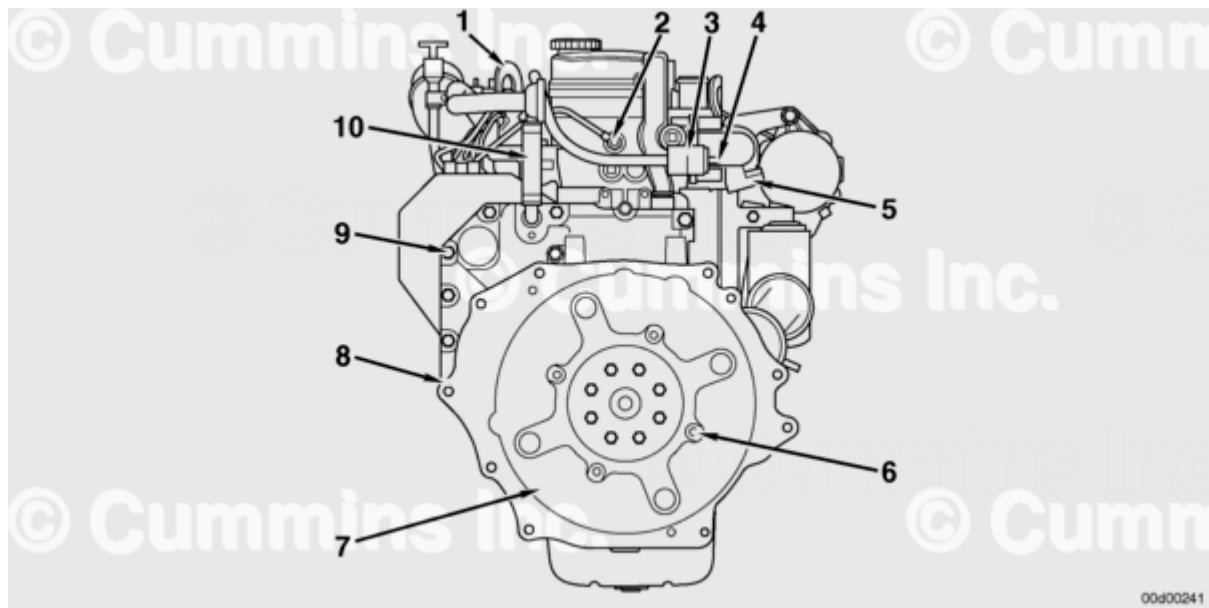
Front View - B4.5 RGT

1. Engine air inlet
2. Lubricating oil dipstick
3. Fan drive belt
4. Front pulley
5. Water pump
6. Automatic belt tensioner
7. Alternator
8. Fan pulley
9. Lubricating oil fill.



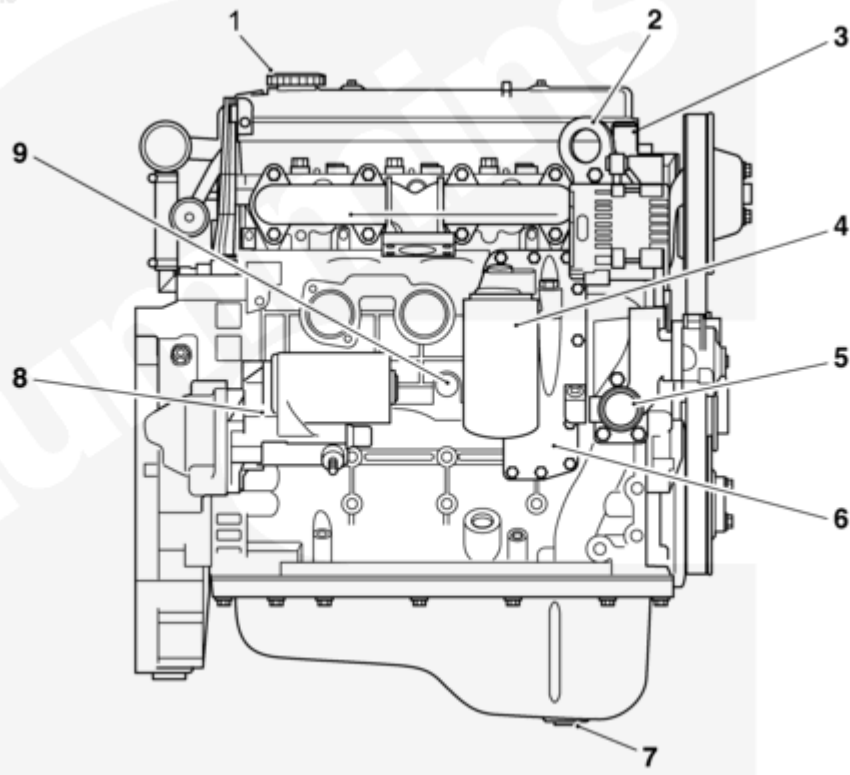
Left Side View - B4.5 RGT

1. Lubricating oil dipstick
2. High pressure fuel lines
3. Rear engine mounting bracket
4. Fuel return line and fuel drain connection
5. Inlet fuel filter
6. Crankcase breather tube
7. Fuel injection pump
8. Engine dataplate
9. Provision for lubricating oil immersion heater
10. Fuel lift pump
11. Magnetic pickup location (3/4-16 UNF)
12. Fuel filter/water separator
13. Intake air preheater (optional)
14. Engine air inlet.



Rear View - B4.5 RGT

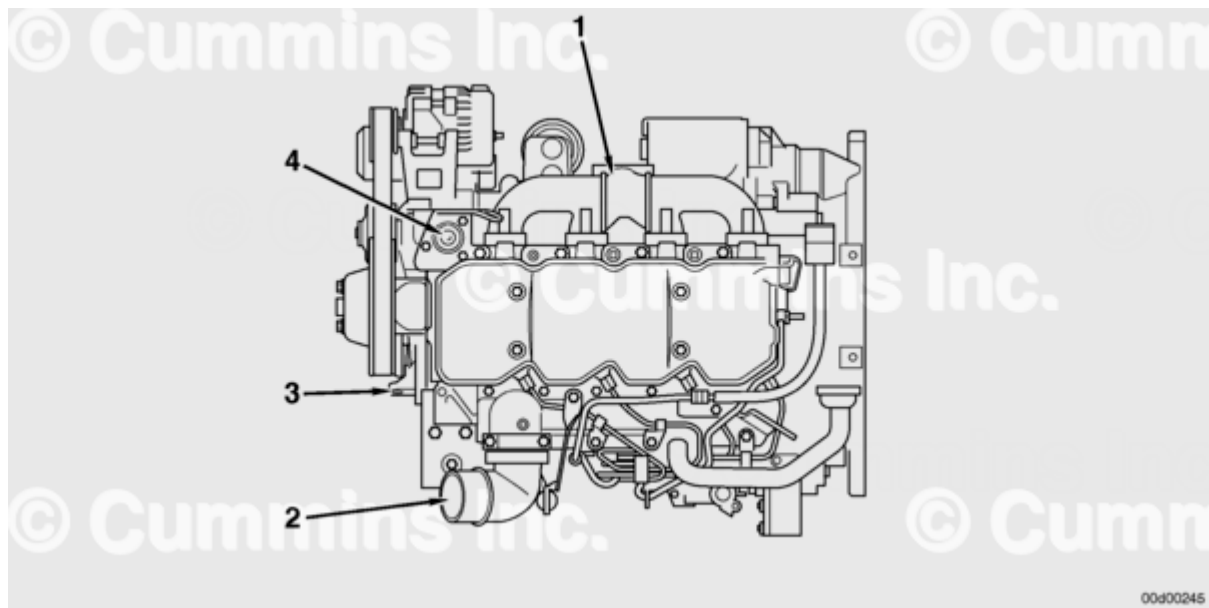
1. Rear engine lifting bracket
2. Fuel return and fuel drain connection
3. Inline fuel filter
4. Fuel Inlet connection
5. Exhaust outlet
6. Flexplate mounting holes
7. Flywheel/flexplate
8. Flywheel housing
9. Rear gear train housing
10. Crankcase breather tube.



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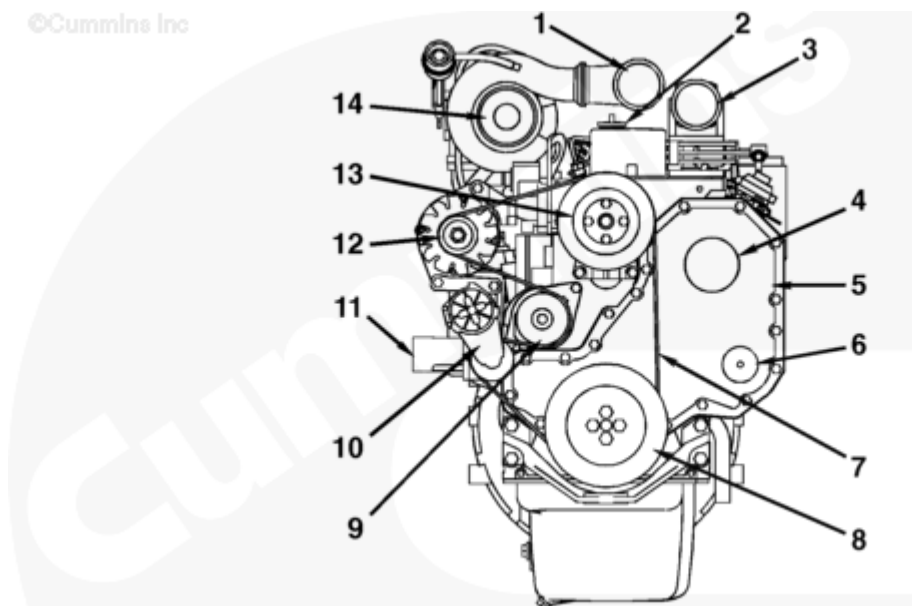
Right Side View - B4.5 RGT

1. Lubricating oil fill
2. Front engine lifting bracket
3. Water outlet
4. Lubricating oil filter
5. Water inlet
6. Lubricating oil cooler
7. Lubricating oil drain
8. Starter motor and solenoid
9. Provision for coolant heater.



Top View - B4.5 RGT

1. Exhaust manifold
2. Engine air inlet
3. Magnetic pickup location (3/4-16 UNF).
4. Water outlet

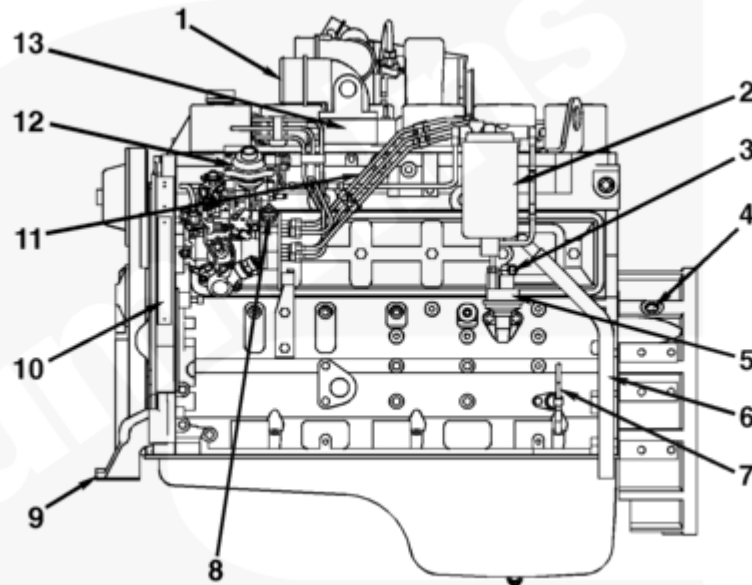


Front View - B5.9 Industrial

1. Turbocharger air outlet
2. Lubricating oil fill
3. Engine air inlet
4. Fuel pump drive cover
5. Front gear cover
6. Accessory drive cover (optional)
7. Fan drive belt

- 8. Vibration damper
- 9. Water pump
- 10. Automatic belt tensioner
- 11. Water inlet
- 12. Alternator
- 13. Fan pulley
- 14. Turbocharger air inlet.

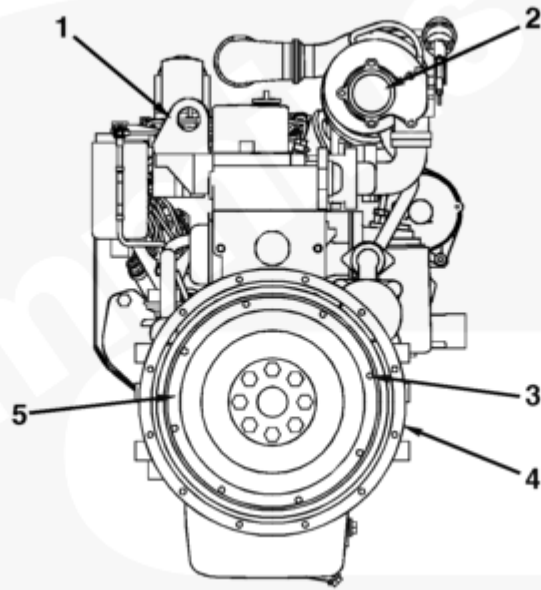
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00900348

Left Side View - B5.9 Industrial

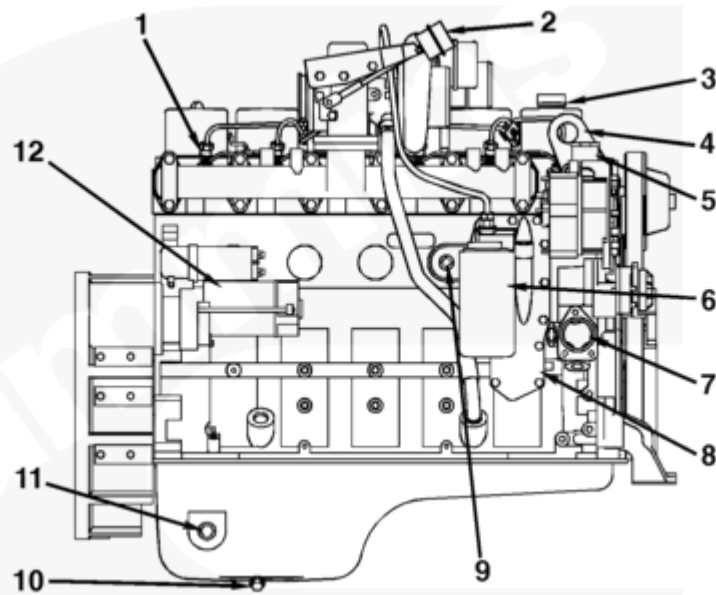
- 1. Engine air inlet
- 2. Fuel filter/water separator
- 3. Fuel inlet connection
- 4. Magnetic pickup location (3/4-16 UNF)
- 5. Fuel lift pump
- 6. Crankcase breather tube
- 7. Lubricating oil dipstick
- 8. Fuel drain connection
- 9. Front engine mounting bracket
- 10. Engine dataplate
- 11. High pressure fuel lines
- 12. Fuel injection pump
- 13. Intake air preheater (optional).



00900349

Rear View - B5.9 Industrial

1. Rear engine lifting bracket
2. Turbocharger exhaust outlet
3. Flexplate mounting holes
4. Flywheel housing
5. Flywheel/flexplate.



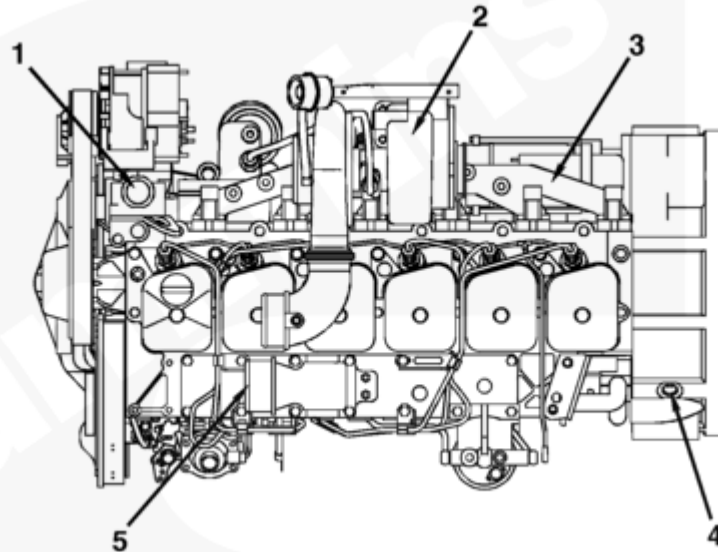
00900350

Right Side View - B5.9 Industrial

1. Fuel injection nozzles
2. Turbocharger wastegate actuator
3. Lubricating oil fill
4. Front engine lifting bracket
5. Water outlet
6. Lubricating oil filter

- 7. Water inlet
- 8. Lubricating oil cooler
- 9. Provision for coolant heater
- 10. Lubricating oil drain
- 11. Provision for lubricating oil immersion heater
- 12. Starter motor and solenoid.

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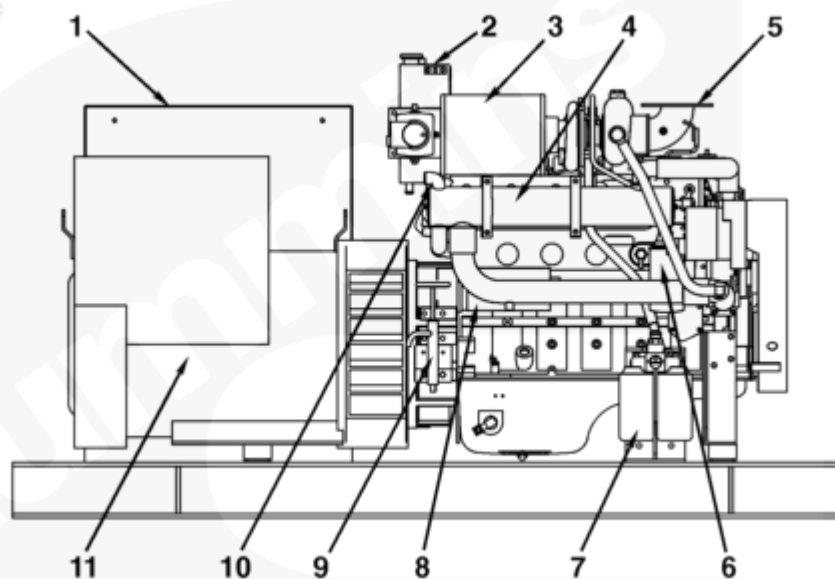


00900362

Top View - B5.9 Industrial

- 1. Water outlet
- 2. Turbocharger
- 3. Exhaust manifold
- 4. Magnetic pickup location (3/4-16 UNF)
- 5. Engine air inlet.

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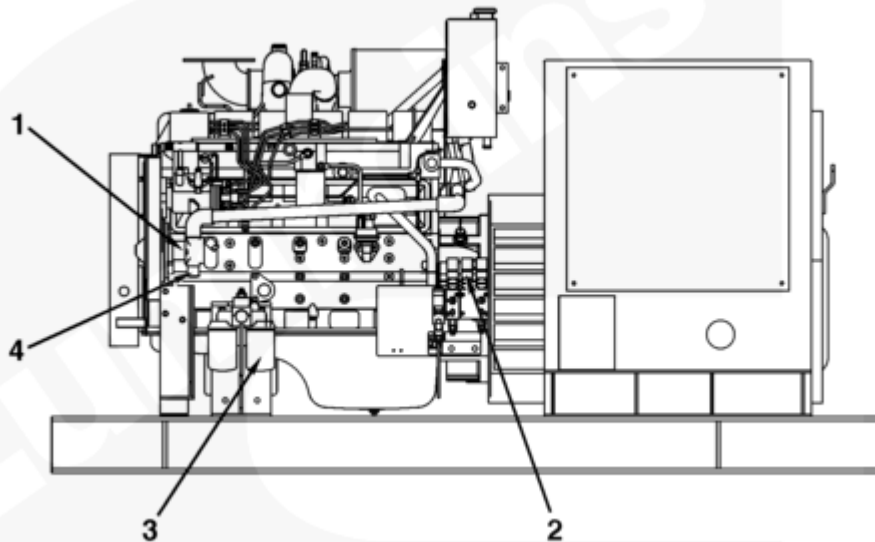


21900008

C Power Heat Exchanger Cooled - Exhaust Side View (B Series)

1. Local Control Panel
2. Coolant Expansion Tank
3. Air Cleaner
4. Heat Exchanger
5. Exhaust Connection
6. Coolant Filter
7. Duplex Lubricating Oil Filter
8. Starter
9. Lubricating Oil Sump Hand Pump
10. Sea Water Outlet Connection
11. Alternator.

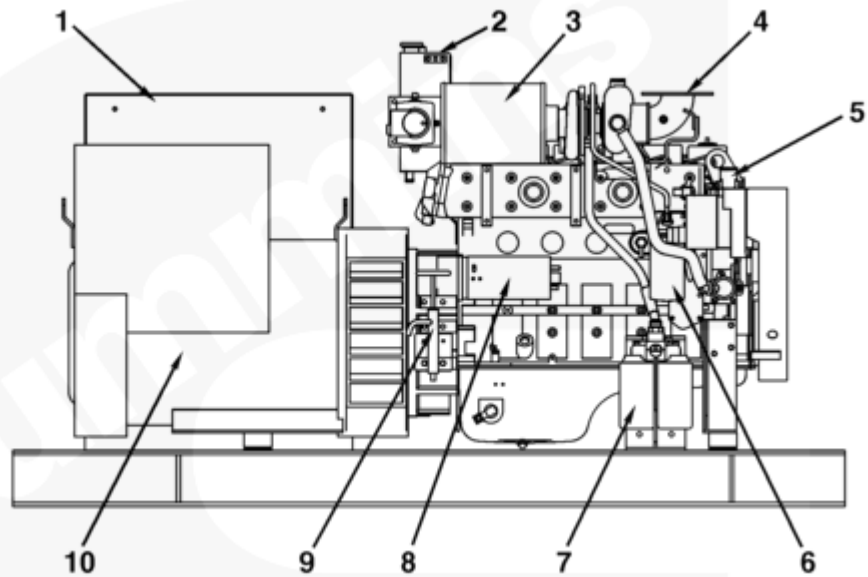
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21900009

C Power Heat Exchanger Cooled - Air Intake View (B Series)

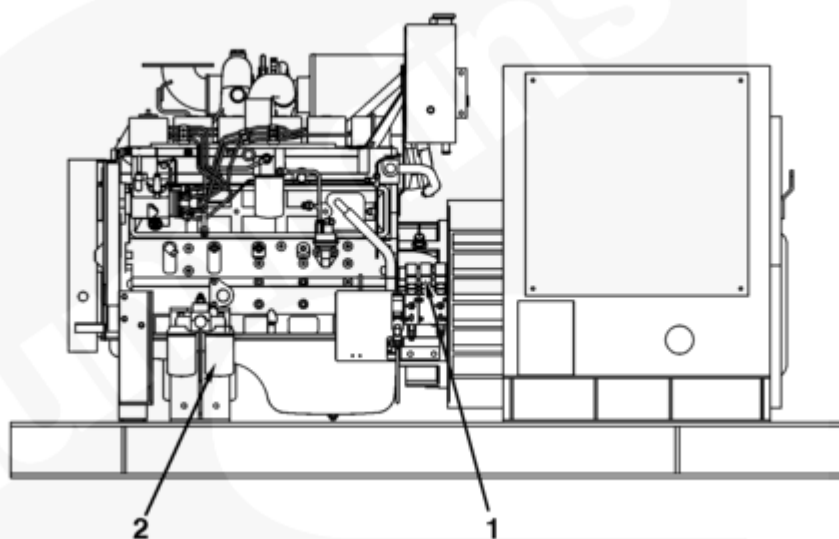
1. Seawater Pump
2. Emergency Shutdown Sensors
3. Fuel Filters
4. Seawater Inlet Connection.



21900010

C Power Keel Cooler Cooled - Exhaust Side View (B Series)

1. Local Control Panel
2. Air Cleaner
3. Coolant Expansion Tank
4. Exhaust Connection
5. Coolant Thermostat Location
6. Coolant Filter
7. Duplex Lubricating Oil Filter
8. Starter
9. Lubricating Oil Sump Hand Pump
10. Alternator.



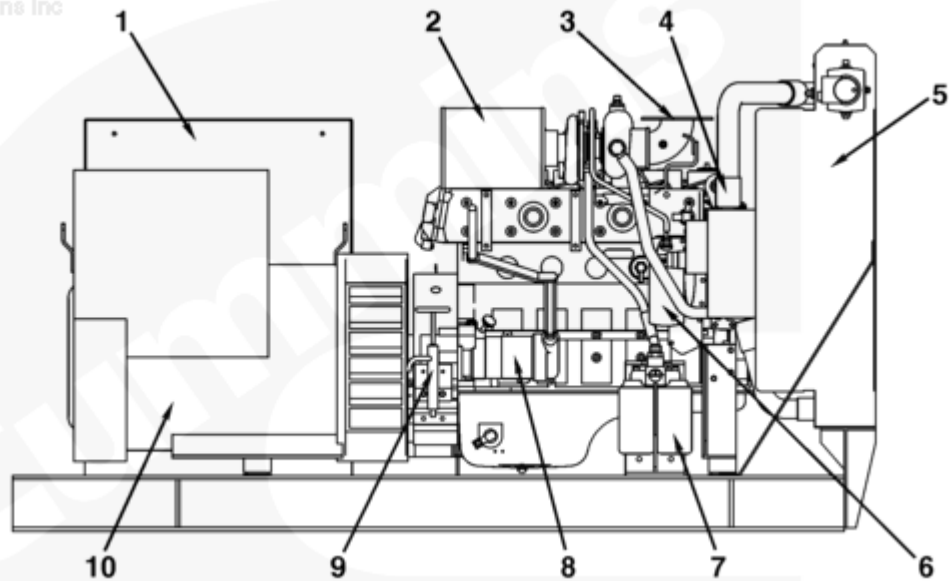
21900011

C Power Keel Cooler Cooled - Air Intake View (B Series)

1. Fuel Filters

2. Emergency Shutdown Sensors.

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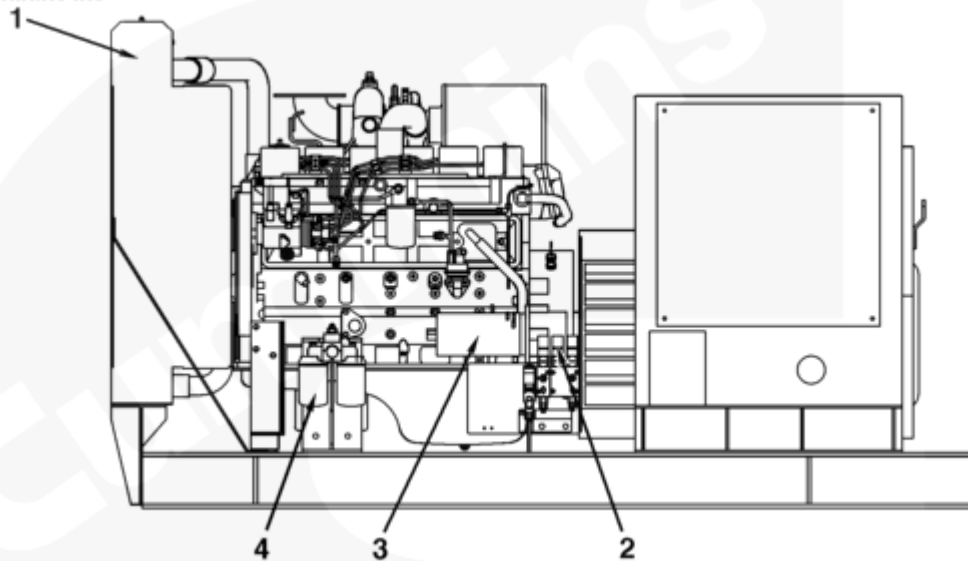


21900012

C Safe Radiator Cooled - Exhaust Side View (B Series)

1. Local Control Panel
2. Air Cleaner
3. Exhaust Connection
4. Coolant Thermostat Location
5. Radiator Fan Drive Shroud
6. Coolant Filter
7. Duplex Lubricating Oil Filter
8. Spring Starter
9. Lubricating Oil Sump Hand Pump
10. Alternator.

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21900013

C Safe Radiator Cooled - Air Intake Side View (B Series)

1. Radiator and Expansion Tank
2. Emergency Shutdown Sensors
3. Starter
4. Duplex Fuel Filters.

Last Modified: 07-Mar-2007
